



Indian Agriculture

M A R W A

A S H R A F



Agenda

Introduction

Import Data

Transformation Data

Data Modeling

Visualization





Introduction

District-wise agricultural data for 20 major crops in India, including area, yield, and production, are provided, enabling detailed analysis of productivity trends. The dataset encompasses both traditional crop varieties and High Yielding Varieties (HYVs), offering insights into modern agricultural practices. With detailed documentation, it ensures clarity and accuracy in analysis. This valuable resource aids policymakers, researchers, and stakeholders in understanding localized agricultural dynamics and optimizing interventions. It serves as a comprehensive tool for evidence-based decision-making and strategic planning in India's agricultural sector.

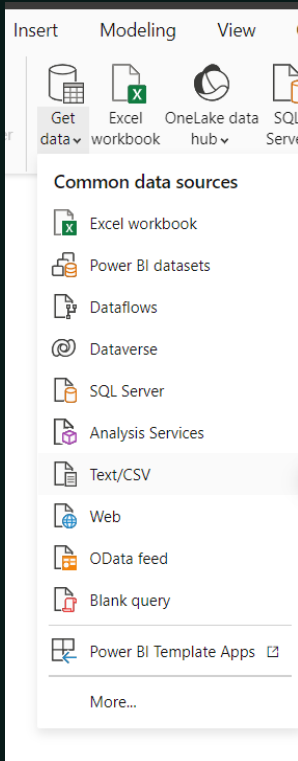


Import Data

POWER BI

Data Overview

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB
1	Dist Code	Year	State Cod	State Name	Dist Name	RICE ARE	RICE PRC	RICE YIEL	WHEAT A	WHEAT P	WHEAT Y	KHARIF S	KHARIF S	KHARIF S	RABI SOR	RABI SOR	RABI SOR	SORGHU	SORGHU	SORGHU	PEARL MI	PEARL MI	PEARL MI	MAIZE AR	MAIZE PR	MAIZE YIE	FINGER M	FINGER M
2	1	1966	14	Chhattisgarh	Durg	548	185	337.59	44	20	454.55	0.6	0.4	666.67	0	0	0	0.6	0.4	666.67	0	0	0	3	2	666.67	0.8	0.2
3	1	1967	14	Chhattisgarh	Durg	547	409	747.71	50	26	520	1.1	0.9	818.18	0	0	0	1.1	0.9	818.18	0	0	0	3	3	1000	0.9	0.2
4	1	1968	14	Chhattisgarh	Durg	556.3	468	841.27	53.7	30	558.66	0.5	0.4	800	0	0	0	0.5	0.4	800	0	0	0	2.8	2	714.29	0.8	0.2
5	1	1969	14	Chhattisgarh	Durg	563.4	400.8	711.4	49.4	26.5	536.44	0.8	0.6	750	0	0	0	0.8	0.6	750	0	0	0	2.7	2.3	851.85	0.8	0.2
6	1	1970	14	Chhattisgarh	Durg	571.6	473.6	828.55	44.2	29	656.11	0.9	0.6	666.67	0	0	0	0.9	0.6	666.67	0	0	0	2.5	3.3	1320	0.8	0.2
7	1	1971	14	Chhattisgarh	Durg	581.8	412.9	709.69	44.4	25.8	581.08	0.3	0.2	666.67	0	0	0	0.3	0.2	666.67	0	0	0	2.7	3.1	1148.15	0.9	0.2
8	1	1972	14	Chhattisgarh	Durg	582.2	381	654.41	39.6	20.6	520.2	0.3	0.3	1000	0	0	0	0.3	0.3	1000	0	0	0	2.8	3.2	1142.86	0.8	0.1
9	1	1973	14	Chhattisgarh	Durg	600	471.9	786.5	37.3	18.6	498.66	0.2	0.2	1000	0	0	0	0.2	0.2	1000	0	0	0	2.9	2.7	931.03	0.8	0.2
10	1	1974	14	Chhattisgarh	Durg	587.4	219	372.83	36.5	22.4	613.7	0.5	0.4	800	0	0	0	0.5	0.4	800	0	0	0	2.9	2.9	1000	0.8	0.3
11	1	1975	14	Chhattisgarh	Durg	598.3	454	758.82	49.2	27.8	565.04	0.2	0.2	1000	0	0	0	0.2	0.2	1000	0	0	0	2.9	2.9	1000	0.8	0.2
12	1	1976	14	Chhattisgarh	Durg	593.6	327.1	551.04	46.9	10	213.22	0.1	0.1	1000	0	0	0	0.1	0.1	1000	0	0	0	3	2	666.67	0.8	0.2
13	1	1977	14	Chhattisgarh	Durg	600.7	572.4	952.89	53.1	27.1	510.36	0.1	0.1	1000	0	0	0	0.1	0.1	1000	0	0	0	3.4	2.8	823.53	0.8	0.2
14	1	1978	14	Chhattisgarh	Durg	612.5	362.2	591.35	48.7	25.6	525.67	0.1	0.1	1000	0	0	0	0.1	0.1	1000	0	0	0	3.4	2.9	852.94	0.7	0.2
15	1	1979	14	Chhattisgarh	Durg	616.8	330.6	535.99	44.6	17.8	399.1	0.5	0.5	1000	0	0	0	0.5	0.5	1000	0	0	0	3	3.6	1200	0.7	0.2
16	1	1980	14	Chhattisgarh	Durg	634.9	515.6	812.1	44.1	33.6	761.9	0.2	0.2	1000	0	0	0	0.2	0.2	1000	0	0	0	3	2.8	933.33	0.6	0.2
17	1	1981	14	Chhattisgarh	Durg	630	506.9	804.6	41.5	23.6	568.67	0.2	0.2	1000	0	0	0	0.2	0.2	1000	0	0	0	3	3	1000	0.7	0.2
18	1	1982	14	Chhattisgarh	Durg	627.9	513.3	817.49	41.1	23.9	581.51	0.1	0.1	1000	0	0	0	0.1	0.1	1000	0	0	0	3	3.1	1033.33	0.6	0.2
19	1	1983	14	Chhattisgarh	Durg	626.7	711	1134.51	39.9	20.6	516.29	0.1	0.1	1000	0	0	0	0.1	0.1	1000	0	0	0	3	3.7	1233.33	0.7	0.2
20	1	1984	14	Chhattisgarh	Durg	632.2	563.8	891.81	40.5	19.9	491.36	0.3	0.3	1000	0	0	0	0.3	0.3	1000	0	0	0	3.6	4.1	1138.89	0.6	0.2
21	1	1985	14	Chhattisgarh	Durg	630.8	699.8	1109.38	39.4	21	532.99	0.3	0.2	666.67	0	0	0	0.3	0.2	666.67	0	0	0	3.8	3	789.47	0.5	0.1
22	1	1986	14	Chhattisgarh	Durg	643	525	816.49	37	24	648.65	0.2	0.1	500	0	0	0	0.2	0.1	500	0	0	0	4	3	750	0.6	0.1
23	1	1987	14	Chhattisgarh	Durg	648	523	807.1	43	23	534.88	0.3	0.2	666.67	0	0	0	0.3	0.2	666.67	0	0	0	4	3	750	0.4	0.1
24	1	1988	14	Chhattisgarh	Durg	652.7	549.7	842.19	43.7	20.2	462.24	0.1	0.1	1000	0	0	0	0.1	0.1	1000	0	0	0	3.8	4	1052.63	0.4	0.1
25	1	1989	14	Chhattisgarh	Durg	660.2	457.3	692.67	43.8	22.7	518.26	0.1	0.1	1000	0	0	0	0.1	0.1	1000	0	0	0	3.7	4.8	1297.3	0.4	0.1
26	1	1990	14	Chhattisgarh	Durg	682.9	806.8	1181.43	36.2	24.8	685.08	0.5	0.4	800	0	0	0	0.5	0.4	800	0	0	0	4	3.8	950	0	0
27	1	1991	14	Chhattisgarh	Durg	680.8	773.6	1136.31	34.8	21.6	620.69	2.3	1.8	782.61	0	0	0	2.3	1.8	782.61	0	0	0	3.8	3.5	921.05	0	0
28	1	1992	14	Chhattisgarh	Durg	688	777.3	1129.8	56.1	20.5	365.42	1.1	1	909.09	0	0	0	1.1	1	909.09	0	0	0	4.4	6	1363.64	0	0
29	1	1993	14	Chhattisgarh	Durg	703.6	876.7	1246.02	33.8	23.8	704.14	0.2	0.2	1000	2.4	3.4	1416.67	2.6	3.6	1384.62	0.01	0.01	1000	3.8	4.2	1105.26	0	0
30	1	1994	14	Chhattisgarh	Durg	726.8	967.9	1331.73	36.5	29.4	805.48	0.3	0.2	666.67	0	0	0	0.3	0.2	666.67	0	0	0	4.8	4.3	895.83	0	0
31	1	1995	14	Chhattisgarh	Durg	723.3	1052	1454.44	37.1	26.5	714.29	0.1	0.1	1000	0	0	0	0.1	0.1	1000	0	0	0	4.2	5.6	1333.33	0	0
32	1	1996	14	Chhattisgarh	Durg	710.9	917.2	1290.2	36.2	29.1	803.87	0.3	0.2	666.67	0	0	0	0.3	0.2	666.67	0.3	0.3	1000	3.2	4.1	1281.25	0	0
33	1	1997	14	Chhattisgarh	Durg	735.7	837.5	1138.37	35.6	18.3	514.04	1.21	0.1	82.64	0.02	0	0	1.24	0.1	80.65	0	0	0	3.5	3.2	914.29	0.26	0
34	1	1998	14	Chhattisgarh	Durg	786.51	684.62	870.45	39.96	24.84	621.62	0.27	0.27	1000	0	0	0	0.27	0.27	1000	0	0	0	5.61	5.68	1012.65	0	0
35	1	1999	14	Chhattisgarh	Durg	769.79	963.04	1251.04	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3.81	4.98	1307.44	0.25	0.07
36	1	2000	14	Chhattisgarh	Durg	748.76	467.89	624.89	29.9	17.59	588.19	0.42	0.23	558.46	0	0	0	0.42	0.23	558.46	0.03	0.02	631	3.94	3.12	792.52	0.23	0.07
37	1	2001	14	Chhattisgarh	Durg	756.26	978.69	1294.13	34.33	24.11	702.43	0.28	0.25	902.45	0	0	0	0.28	0.25	902.45	0.02	0	0	3.9	2.62	672.21	0.22	0.06
38	1	2002	14	Chhattisgarh	Durg	753.41	484.9	643.6	33.25	25.09	754.44	0.22	0.21	947.81	0	0	0	0.22	0.21	947.81	0.01	0	0	3.71	4.76	1284.3	0.23	0.05
39	1	2003	14	Chhattisgarh	Durg	766.58	1145.82	1494.73	37.99	29.58	778.78	0.17	0.15	872.07	0	0	0	0.17	0.15	872.07	0	0	0	3.4	3.35	983.49	0.24	0.08



Power BI

ICRISAT-District Level Data.csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

Dist Code	Year	State Code	State Name	Dist Name	RICE AREA (1000 ha)	RICE PRODUCTION (1000 tons)	RICE YIELD (Kg per ha)
1	1966	14	Chhattisgarh	Durg	548	185	337.55
1	1967	14	Chhattisgarh	Durg	547	409	747.71
1	1968	14	Chhattisgarh	Durg	556.3	468	841.27
1	1969	14	Chhattisgarh	Durg	563.4	400.8	711.4
1	1970	14	Chhattisgarh	Durg	571.6	473.6	828.55
1	1971	14	Chhattisgarh	Durg	581.8	412.9	709.65
1	1972	14	Chhattisgarh	Durg	582.2	381	654.41
1	1973	14	Chhattisgarh	Durg	600	471.9	786.5
1	1974	14	Chhattisgarh	Durg	587.4	219	372.85
1	1975	14	Chhattisgarh	Durg	598.3	454	758.82
1	1976	14	Chhattisgarh	Durg	593.6	327.1	551.04
1	1977	14	Chhattisgarh	Durg	600.7	572.4	952.85
1	1978	14	Chhattisgarh	Durg	612.5	362.2	591.35
1	1979	14	Chhattisgarh	Durg	616.8	330.6	535.95
1	1980	14	Chhattisgarh	Durg	634.9	515.6	812.1
1	1981	14	Chhattisgarh	Durg	630	506.9	804.6
1	1982	14	Chhattisgarh	Durg	627.9	513.3	817.45
1	1983	14	Chhattisgarh	Durg	626.7	711	1134.51
1	1984	14	Chhattisgarh	Durg	632.2	563.8	891.81
1	1985	14	Chhattisgarh	Durg	630.8	699.8	1109.38

Extract Table Using Examples | Load | Transform Data | Cancel

6

Import Data From CSV to Power BI

	1 ² 3 Dist Code	1 ² 3 Year	1 ² 3 State Code	A ^B C State Name	A ^B C Dist Name	1.2 RICE AREA (1000 ha)	1
1	160	1966	10	Rajasthan	Swami Madhopur		
2	240	1966	16	Uttarakhand	Chamoli		
3	106	1966	7	Maharashtra	Solapur		
4	911	1966	2	Bihar	Shahabad (now part of Bhojpur district)	438.9	
5	244	1966	16	Uttarakhand	Dehradun	12.	
6	5	1966	14	Chhattisgarh	Raigarh	33	
7	153	1966	10	Rajasthan	Jalore		
8	137	1966	3	Gujarat	Surat	5	
9	25	1966	6	Madhya Pradesh	Datia		
10	908	1966	2	Bihar	Mungair	115.5	
11	96	1966	7	Maharashtra	Thane	15	
12	652	1966	19	Kerala	Kannur	92.8	
13	627	1966	18	Himachal Pradesh	Chamba	3.1	
14	51	1966	1	Andhra Pradesh	Kurnool	9	
15	108	1966	7	Maharashtra	Aurangabad		
16	195	1966	12	Uttar Pradesh	Mathura	1.	
17	239	1966	16	Uttarakhand	Pithorgarh		
18	135	1966	3	Gujarat	Rajkot		
19	703	1966	8	Orissa	Cuttack	626.9	
20	231	1966	12	Uttar Pradesh	Faizabad	145.	
21	189	1966	12	Uttar Pradesh	Saharanpur	89.	
22	184	1966	4	Haryana	Karnal	125.	
23	223	1966	12	Uttar Pradesh	Basti	315.	
24							

Transform Data into Power Query



Transformation Data

POWER BI





AB _C Key
310 distinct, 0 unique
14-1
7-118
11-92
7-119
7-120
3-121
11-91
3-122
3-123
11-90
7-113
7-95
7-112
7-106
7-102
7-100
7-117
11-93
7-116
7-115
4-182
4-183
4-184



Add Custom Columns Total Area , Production , Yield

Custom Column

Add a column that is computed from the other columns.

New column name

Custom column formula

Year

Key

State Name

Dist Name

Rice Area (1000 ha)

Rice Production (1000 tons)

Rice Yield (Kg per ha)

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK

Cancel

Custom Column

Add a column that is computed from the other columns.

New column name

Custom column formula

Year

Key

State Name

Dist Name

Rice Area (1000 ha)

Rice Production (1000 tons)

Rice Yield (Kg per ha)

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK

Cancel

Custom Column

Add a column that is computed from the other columns.

New column name

Custom column formula

Year

Key

State Name

Dist Name

Rice Area (1000 ha)

Rice Production (1000 tons)

Rice Yield (Kg per ha)

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK

Cancel



Add Custom Columns Average Area , Production , Yield

Custom Column

Add a column that is computed from the other columns.

New column name

AVG Area

Custom column formula

= List.Average({
 [#"Rice Area (1000 ha)"],
 [#"Wheat Area (1000 ha)"],
 [#"Sorghum Area (1000 ha)"],
 [#"Pearl Millet Area (1000 ha)"],
 [#"Maize Area (1000 ha)"],
 [#"Finger Millet Area (1000 ha)"],
 [#"Barley Area (1000 ha)"],
 [#"Chickpea Area (1000 ha)"],
 [#"Pigeonpea Area (1000 ha)"],
 [#"Minor Pulses Area (1000 ha)"],
})

Available columns

Year

Key

State Name

Dist Name

Rice Area (1000 ha)

Rice Production (1000 tons)

Rice Yield (Kg per ha)

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK

Cancel

Custom Column

Add a column that is computed from the other columns.

New column name

AVG Production

Custom column formula

= List.Average({
 [#"Rice Production (1000 tons)"],
 [#"Wheat Production (1000 tons)"],
 [#"Sorghum Production (1000 tons)"],
 [#"Pearl Millet Production (1000 tons)"],
 [#"Maize Production (1000 tons)"],
 [#"Finger Millet Production (1000 tons)"],
 [#"Barley Production (1000 tons)"],
 [#"Chickpea Production (1000 tons)"],
 [#"Pigeonpea Production (1000 tons)"],
 [#"Minor Pulses Production (1000 tons)"],
})

Available columns

Year

Key

State Name

Dist Name

Rice Area (1000 ha)

Rice Production (1000 tons)

Rice Yield (Kg per ha)

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK

Cancel

Custom Column

Add a column that is computed from the other columns.

New column name

AVG Yield

Custom column formula

= List.Average({
 [#"Rice Yield (Kg per ha)"],
 [#"Wheat Yield (Kg per ha)"],
 [#"Sorghum Yield (Kg per ha)"],
 [#"Pearl Millet Yield (Kg per ha)"],
 [#"Maize Yield (Kg per ha)"],
 [#"Finger Millet Yield (Kg per ha)"],
 [#"Barley Yield (Kg per ha)"],
 [#"Chickpea Yield (Kg per ha)"],
 [#"Pigeonpea Yield (Kg per ha)"],
 [#"Minor Pulses Yield (Kg per ha)"],
})

Available columns

Year

Key

State Name

Dist Name

Rice Area (1000 ha)

Rice Production (1000 tons)

Rice Yield (Kg per ha)

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK

Cancel

	1 ² ₃ Index	A ^B _C Dist Name
		
	311 distinct, 311 unique	311 distinct, 311 unique
1		1 Durg
2		2 Bastar
3		3 Raipur
4		4 Bilaspur
5		5 Raigarh
6		6 Surguja
7		7 Jabalpur
8		8 Balaghat
9		9 Chhindwara
10		10 Narsinghpur
11		11 Seoni / Shivani
12		12 Mandla
13		13 Sagar
14		14 Damoh
15		15 Tikamgarh
16		16 Chhatarpur
17		17 Panna
18		18 Rewa
19		19 Sidhi
20		20 Satna
21		21 Shahdol
22		22 Gwalior
23		23 Shivpuri
24		24 Guna

	1 ² ₃ Index	A ^B _C State Name
		
	20 distinct, 20 unique	20 distinct, 20 unique
1		1 Chhattisgarh
2		2 Madhya Pradesh
3		3 Andhra Pradesh
4		4 Telangana
5		5 Karnataka
6		6 Tamil Nadu
7		7 Maharashtra
8		8 Gujarat
9		9 Rajasthan
10		10 Punjab
11		11 Haryana
12		12 Uttar Pradesh
13		13 Uttarakhand
14		14 Assam
15		15 Himachal Pradesh
16		16 Kerala
17		17 Orissa
18		18 West Bengal
19		19 Bihar
20		20 Jharkhand

Create Table State With Distinct State Name and Table District with Distinct District Name



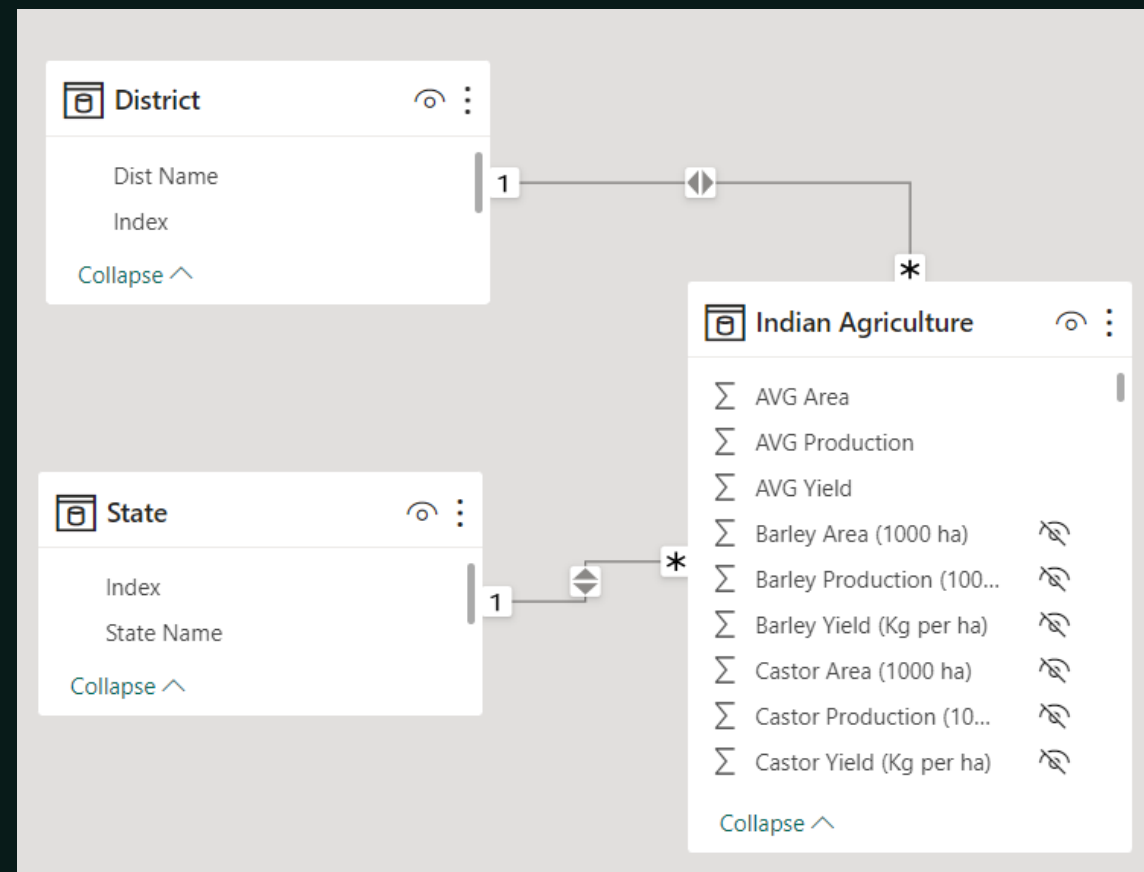
Data Modeling

POWER BI





Data Model of Indian Agriculture Dataset



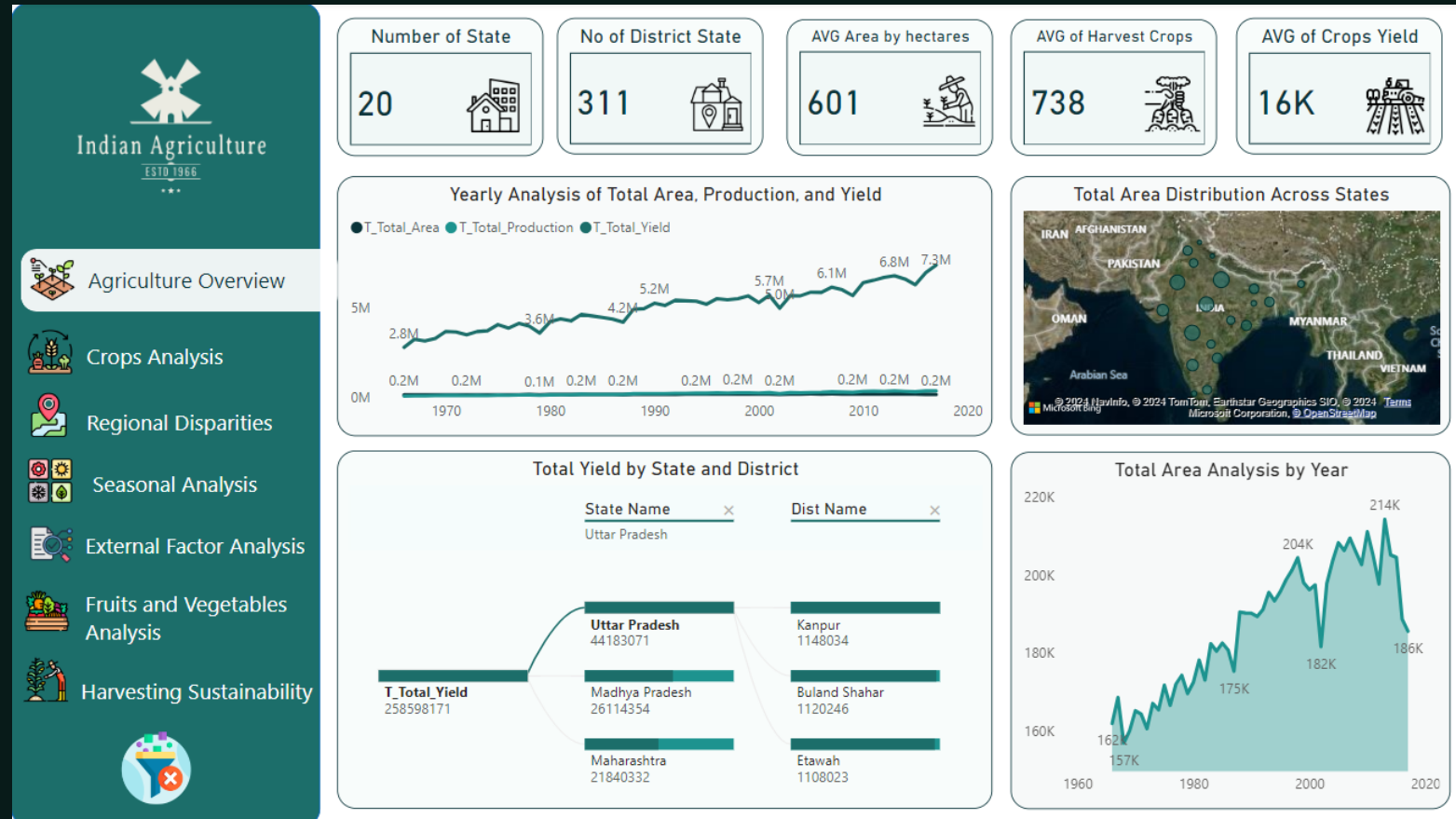


Data Visualization

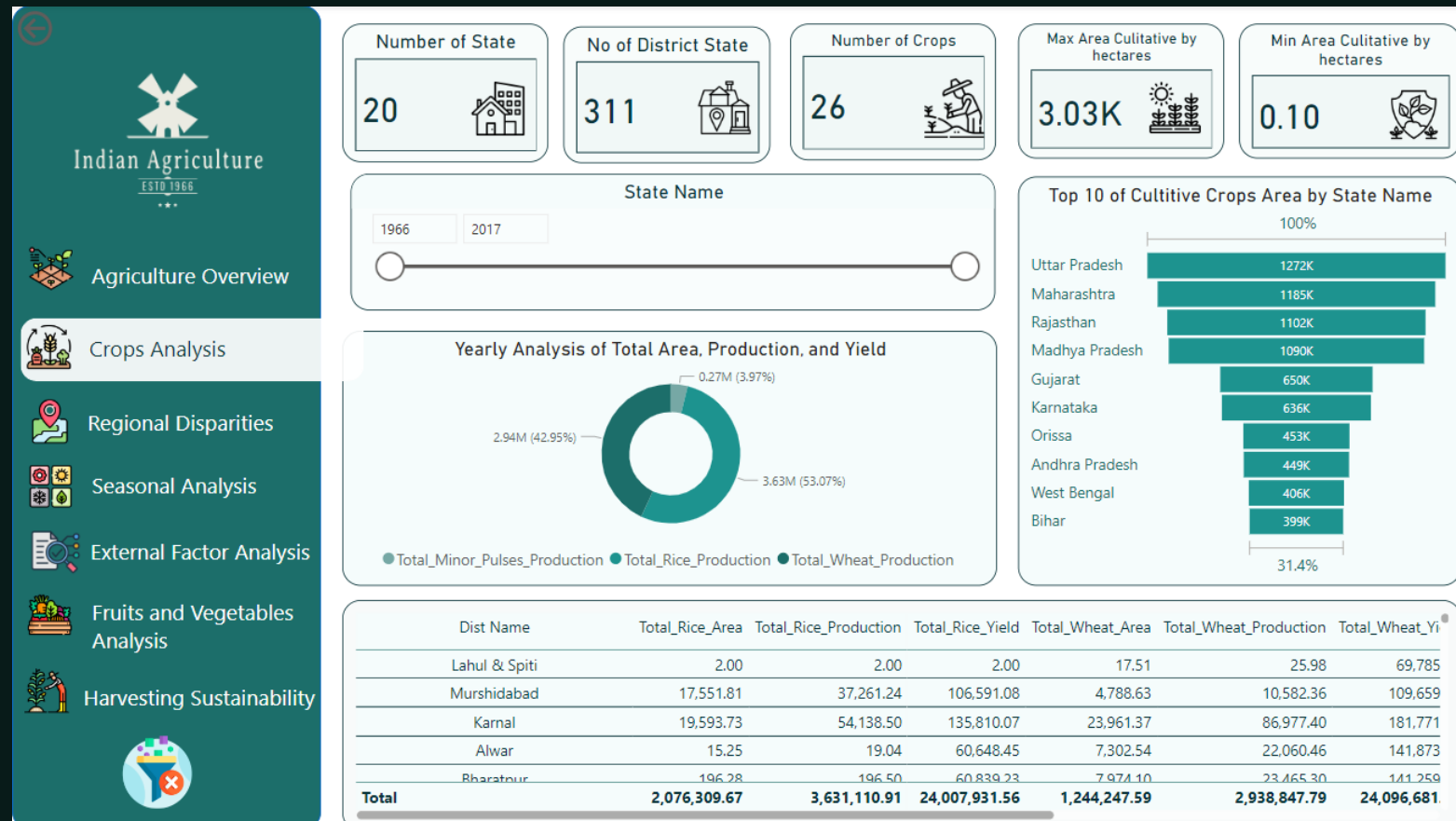
POWER BI



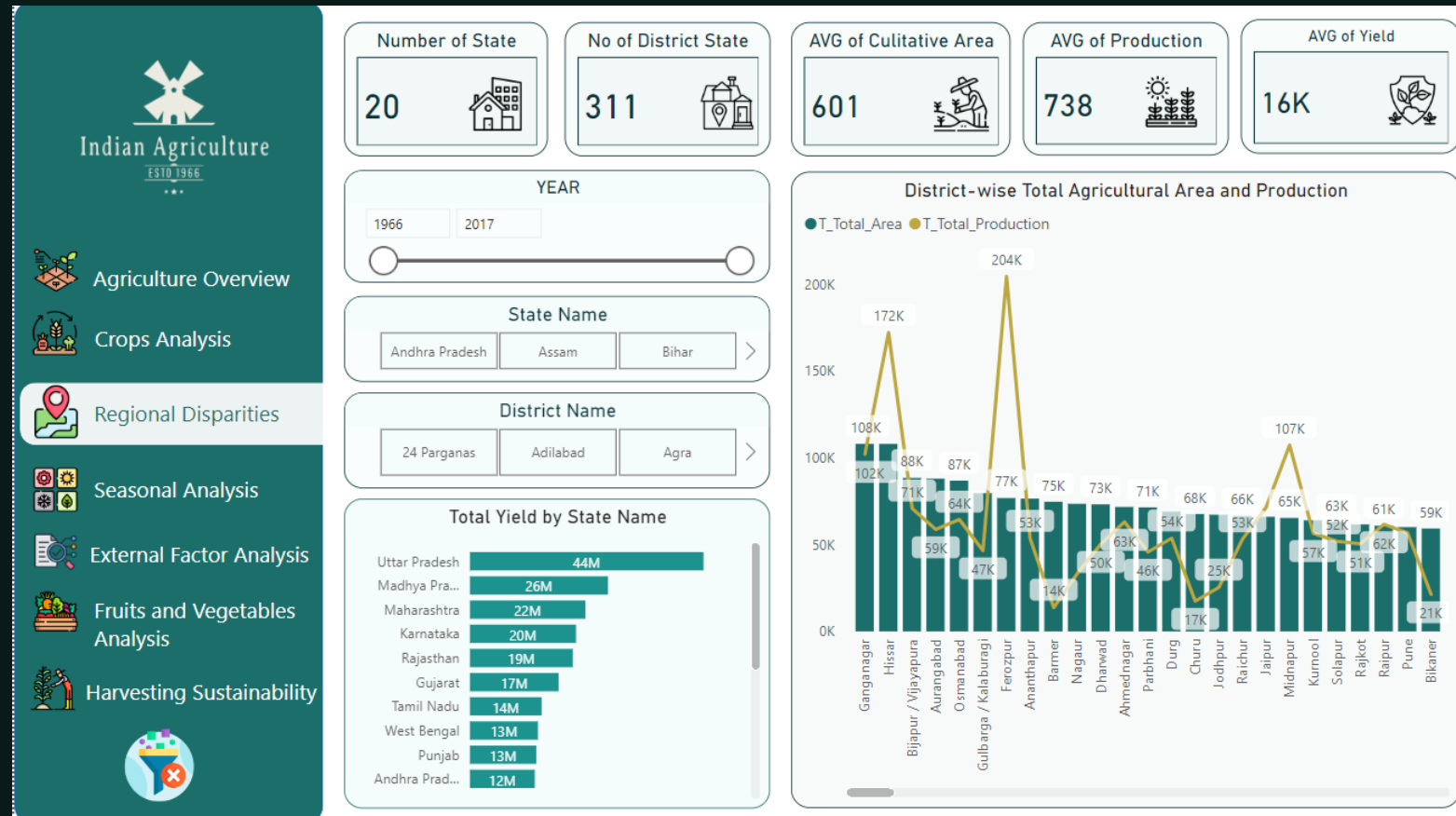
Agriculture Overview



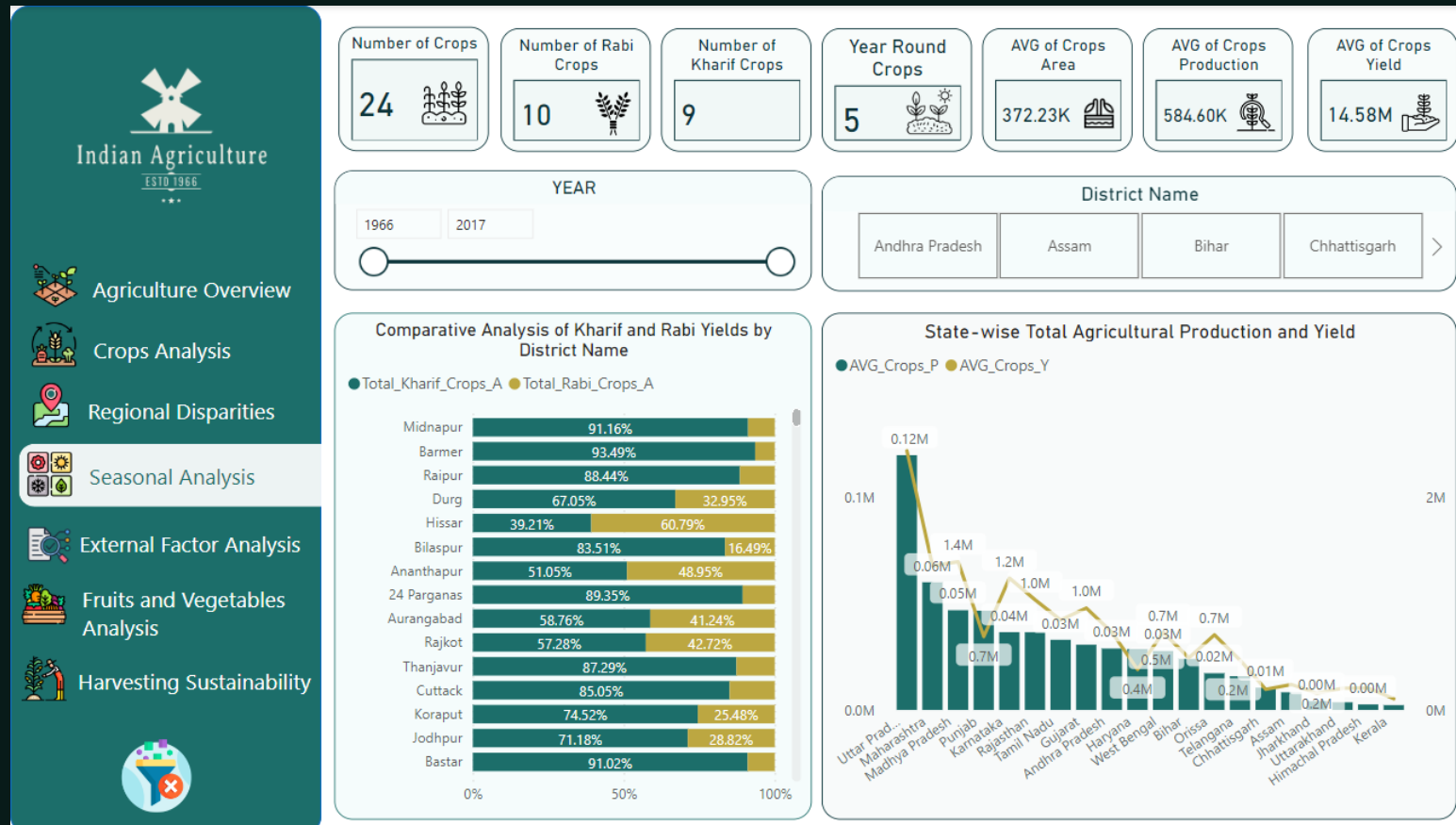
Crops Analysis



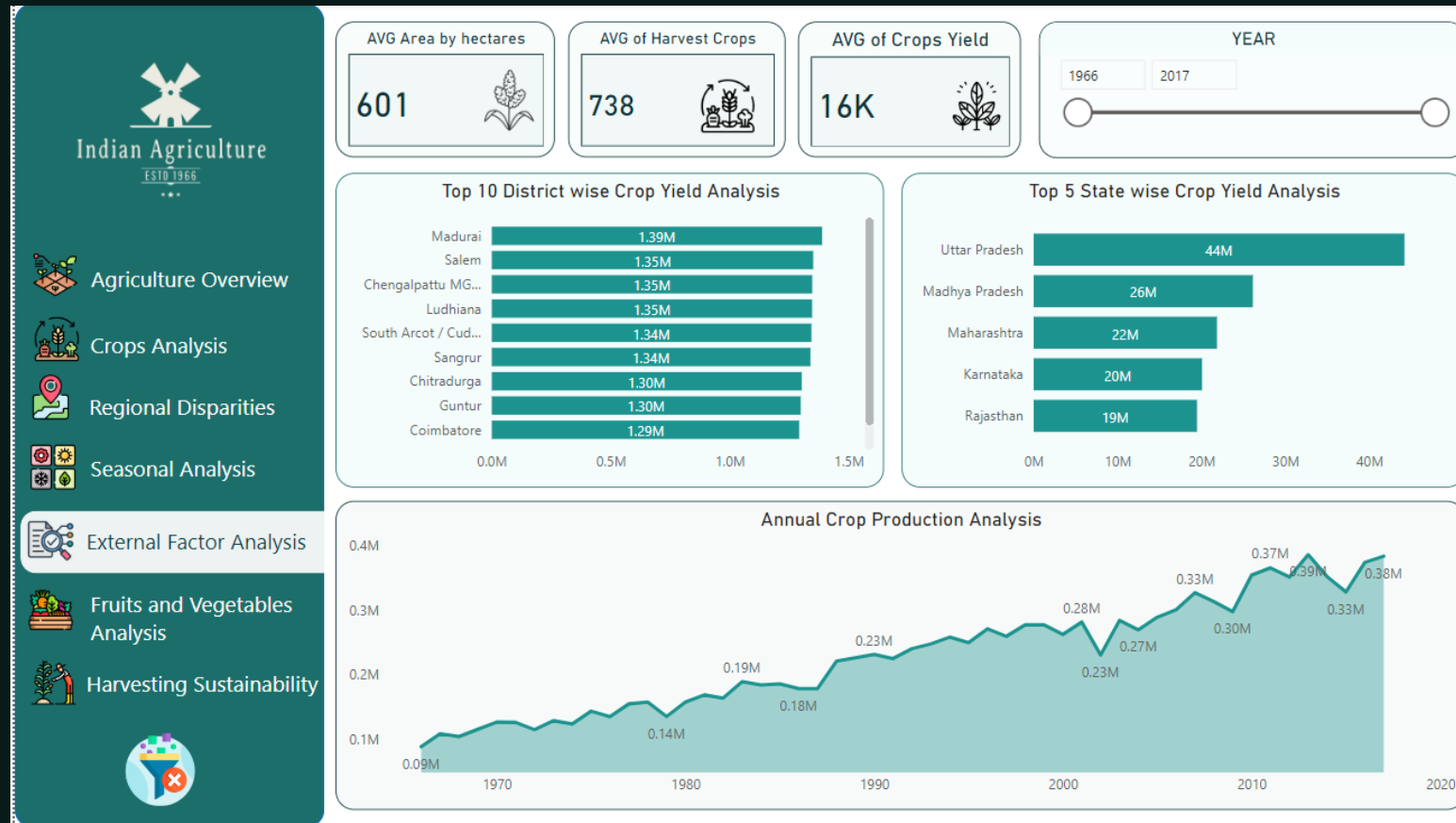
Regional Disparities



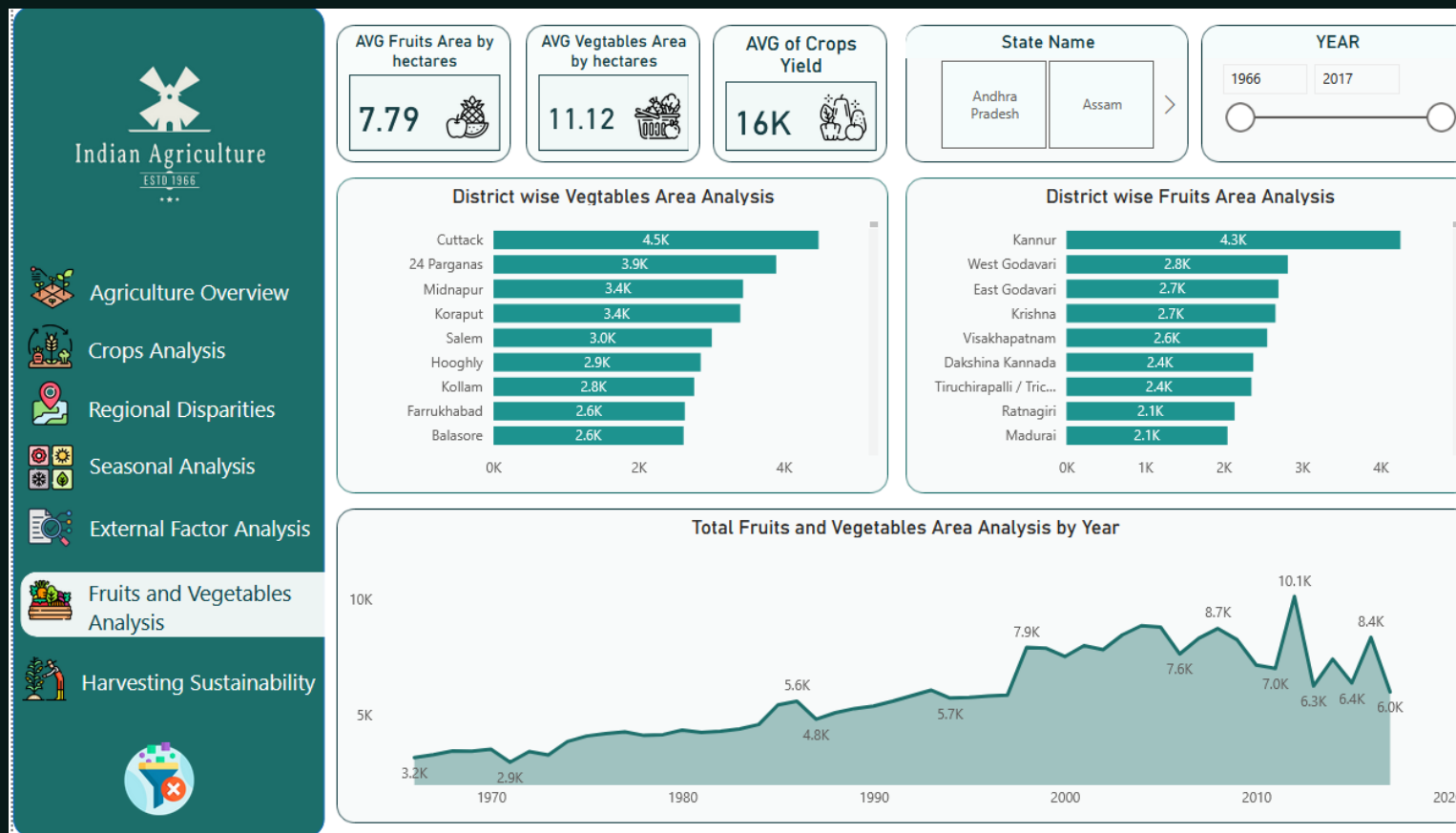
Seasonal Analysis



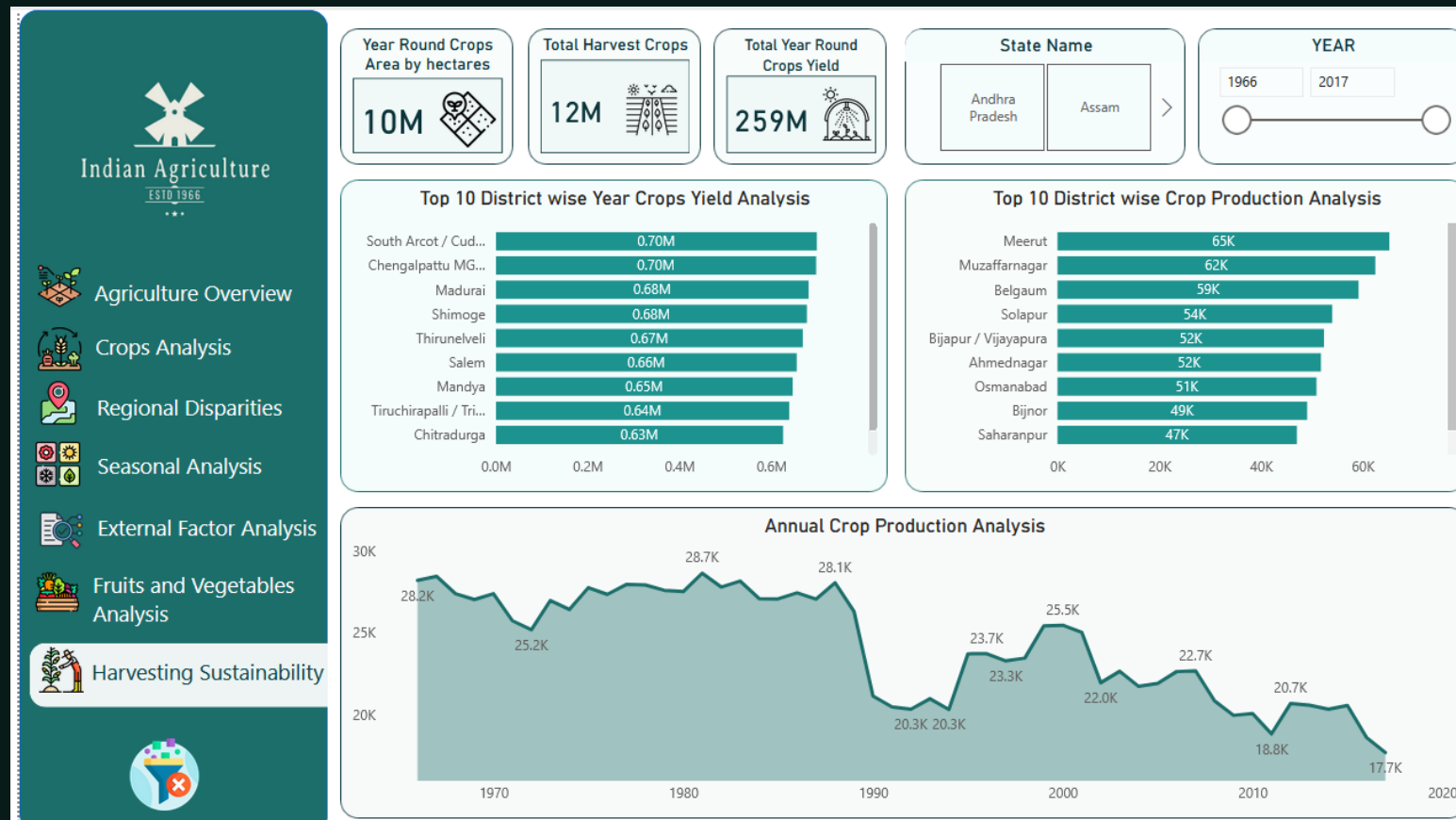
External Factor Analysis



Fruits and Vegetables Analysis



Harvesting Sustainability





Summary

"Through advanced data analysis techniques, valuable insights can be gleaned from agricultural data, enabling optimization of crop yields, resource allocation, and pest management strategies. By leveraging data-driven approaches, farmers can make informed decisions to enhance productivity, sustainability, and resilience in their operations."





Thank you

Marwa Ashraf

marwaashraf5814@gmail.com

<https://www.novopro.com/project/indian-agriculture>

